Context Window Components: Vision Language Model Analysis

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Project: LMM_POC - Vision Language Model Document Extraction
Models: Llama-3.2-11B-Vision, InternVL3-8B

Executive Summary

This document provides a complete breakdown of how the context window is consumed in vision-language model inference. The analysis shows that the current implementation uses only **8-14% of available context**, providing significant safety margins for production deployment.

Key Findings:

- **InternVL3-8B**: Uses 4,724 tokens of 32,768 (14.4% utilization)
- **Llama-3.2-11B**: Uses 10,796 tokens of 128,000 (8.4% utilization)
- **Safety margin**: 3.8x to 9.9x headroom in worst-case scenarios
- **Stateless design**: No conversation history accumulation keeps usage constant

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1 Context Window Components Overview

The context window is consumed by **five main components** during inference:

1.1 Component Summary

1. **System Prompt / Instructions (Optional)**
   - Pre-instructions to guide model behavior
   - **Current implementation**: Not used (0 tokens)
   - **Typical range**: 0-500 tokens
2. **User Prompt (Extraction Instructions)**
   - Document-aware extraction prompt
   - **Current implementation**: 500-800 tokens
   - Varies by document type (invoice, receipt, bank statement)
3. **Image(s) (Visual Tokens)**
   - Encoded representation of document image
   - **InternVL3-8B**: 1,536-5,120 tokens (6-20 tiles)
   - **Llama-3.2-11B**: 4,000-8,000 tokens (high-res documents)
4. **Conversation History (Optional)**
   - Previous turns in multi-turn conversations
   - **Current implementation**: 0 tokens (stateless processing)
   - **If enabled**: Would grow with each turn
5. **Model Response (Generated Output)**
   - The extracted data being generated
   - **InternVL3**: 1,024 tokens max (max_new_tokens)
   - **Llama**: 4,096 tokens max (max_new_tokens)

2 Detailed Component Breakdown

2.1 User Prompt Structure

Prompt Component	Tokens
Task Description	100
Field Definitions	200-400
Output Format	150
Special Rules/Examples	50-150
Total	**500-800**

Table 1: User prompt token breakdown

Document Type	Field Count	Estimated Tokens
Universal	25 fields	700-800
Invoice	14 fields	600-700
Receipt	14 fields	600-700
Bank Statement	7 fields	500-600

Table 2: Prompt tokens by document type

2.2 User Prompt by Document Type

2.3 Image Token Consumption

2.3.1 InternVL3 Image Encoding

Architecture: Dynamic tiling with pixel unshuffle

Process:

1. Image divided into 448×448 pixel tiles
2. Pixel unshuffle reduces tokens to 1/4 of original
3. Each tile = **256 tokens**
4. Variable number of tiles based on aspect ratio

Configuration:

- **Max tiles (8B model)**: 20 tiles (configurable)
- **Worst case**: 20 × 256 = 5,120 tokens
- **Typical document**: 6-12 tiles = 1,536-3,072 tokens

Official Documentation:

> *"InternVL3 applies a pixel unshuffle operation that reduces visual tokens to one-quarter of the original value. Each 448×448 image tile is represented with 256 visual tokens."*

Reference: https://arxiv.org/html/2504.10479v1

2.3.2 Llama 3.2 Vision Image Encoding

Architecture: Fixed encoding based on resolution

Official Measurements:

- **512×512 image**: 1,610 tokens
- **Maximum resolution**: 1120×1120 pixels
- **Scaling**: Approximately linear with pixel count

Scaling Formula:

$$tokens = \frac{(width \times height) \times 1610}{512 \times 512} \tag{1}$$

Official Documentation:

> *"Llama 3.2 VLMs support long context lengths of up to 128K text tokens as well as a single image input at a resolution of 1120 x 1120 pixels. A 512 x 512 image is converted to about 1,610 tokens."*

Reference: https://docs.oracle.com/en-us/iaas/Content/generative-ai/meta-llama-3-2-11b.htm

3 InternVL3-8B Context Consumption

3.1 Complete Breakdown

Component	**Tokens**
INPUTS	
System Prompt	0
User Prompt (extraction)	700
Image (document)	3,000
Total Input	3,700
OUTPUTS	
Conversation History	0
Model Response	1,024
Total Output	1,024
TOTAL USED	**4,724**
REMAINING CAPACITY	**28,044**
UTILIZATION	**14.4%**

Table 3: InternVL3-8B context window usage (Total: 32,768 tokens)

3.2 Worst-Case Scenario

Component	**Typical**	**Worst Case**	**Buffer**
User Prompt	700	800	+100
Image (max tiles)	3,000	5,120	+2,120
Response	1,024	1,024	+0
TOTAL	4,724	6,944	+2,220
Remaining	28,044	25,824	-2,220
Utilization %	14.4%	21.2%	+6.8%

Table 4: InternVL3-8B worst-case analysis

Even in worst case: 78.8% of context window remains unused

4 Llama-3.2-11B-Vision Context Consumption

4.1 Complete Breakdown

4.2 Worst-Case Scenario

Even in worst case: 89.9% of context window remains unused

Component	Tokens
INPUTS	
System Prompt	0
User Prompt (extraction)	700
Image (document)	6,000
Total Input	6,700
OUTPUTS	
Conversation History	0
Model Response	4,096
Total Output	4,096
TOTAL USED	**10,796**
REMAINING CAPACITY	**117,204**
UTILIZATION	**8.4%**

Table 5: Llama-3.2-11B context window usage (Total: 128,000 tokens)

Component	Typical	Worst Case	Buffer
User Prompt	700	800	+100
Image (max resolution)	6,000	8,000	+2,000
Response	4,096	4,096	+0
TOTAL	10,796	12,896	+2,100
Remaining	117,204	115,104	-2,100
Utilization %	8.4%	10.1%	+1.7%

Table 6: Llama-3.2-11B worst-case analysis

5 Image Token Consumption

5.1 Comparison Table

5.2 Example Calculations

5.2.1 InternVL3-8B Examples

```
Document Size: 800×1200 (portrait invoice)

Aspect ratio: 1:1.5
Optimal tiles: 2×3 = 6 tiles
Token count: 6 × 256 = 1,536 tokens
Context %: 1,536 / 32,768 = 4.7%

Document Size: 1600×1200 (landscape statement)

Aspect ratio: 4:3
Optimal tiles: 4×3 = 12 tiles
Token count: 12 × 256 = 3,072 tokens
Context %: 3,072 / 32,768 = 9.4%
```

Model	Context Window	Image (Typical)	Image (Max)	Image % (Typical)	Image % (Max)
InternVL3-8B	32,768	1,536-3,072	5,120	4.7-9.4%	15.6%
Llama-3.2-11B	128,000	4,000-6,000	8,000	3.1-4.7%	6.25%

Table 7: Image token consumption comparison

5.2.2 Llama-3.2-11B Examples

```
Low Resolution: 800×600

Pixels: 480,000
Token count: 480,000 × 6.14 / 1,000 = 2,947 tokens
Context %: 2,947 / 128,000 = 2.3%

High Resolution: 1120×1120 (maximum)

Pixels: 1,254,400
Token count: 1,254,400 × 6.14 / 1,000 = 7,702 tokens
Context %: 7,702 / 128,000 = 6.0%
```

6 Context Window Limits by Provider

6.1 InternVL3-8B Provider Limits

Provider	Theoretical Max	Practical Limit	Restriction
Self-hosted	32,768	32,768	None
Recommended	32,768	16,384	Performance opt.
Deployment	32,768	8,192-16,384	Memory constraints

Table 8: InternVL3-8B context limits by provider

6.2 Llama-3.2-11B-Vision Provider Limits

Provider	Theoretical Max	Actual Limit	% of Max
Meta Specification	128,000	128,000	100%
AWS Bedrock	128,000	128,000	100%
Oracle Cloud	128,000	128,000	100%
Azure Serverless	128,000	**8,000**	**6.25%**

Table 9: Llama-3.2-11B context limits by provider

Critical Issue with Azure:

"Documentation about Llama 3.2 11B Vision Instruct Model says 128K context window but not able to process more than 8k tokens on Azure serverless deployment."

Reference: https://learn.microsoft.com/en-us/answers/questions/2150702/

Implication: Your implementation uses 10,796 tokens, which **exceeds Azure's 8K limit** but works fine on AWS/Oracle/self-hosted.

7 Safety Margins Analysis

7.1 InternVL3-8B Safety Margins

Component	Used	Available	Headroom	Safety Factor
User Prompt	700	32,068	31,368	45.7x
Image (typical)	3,000	29,768	26,768	8.9x
Image (worst)	5,120	27,648	22,528	4.4x
Response	1,024	26,624	25,600	26.0x
TOTAL (typical)	4,724	28,044	23,320	4.9x
TOTAL (worst)	6,944	25,824	18,880	2.7x

Table 10: InternVL3-8B safety margins

7.2 Llama-3.2-11B-Vision Safety Margins

Component	Used	Available	Headroom	Safety Factor
User Prompt	700	127,300	126,600	181.9x
Image (typical)	6,000	122,000	116,000	19.3x
Image (worst)	8,000	120,000	112,000	14.0x
Response	4,096	115,904	111,808	27.3x
TOTAL (typical)	10,796	117,204	106,408	9.9x
TOTAL (worst)	12,896	115,104	102,208	7.9x

Table 11: Llama-3.2-11B safety margins

7.3 Safety Factor Interpretation

- $>$ **5.0x**: Extremely safe, can handle major increases
- **3.0-5.0x**: Very safe, comfortable margin for growth
- **2.0-3.0x**: Safe, adequate buffer for typical variations
- $<$ **2.0x**: Tight, monitor carefully

Current Status:

- InternVL3-8B: Very safe (2.7x - 4.9x safety factor)
- Llama-3.2-11B: Extremely safe (7.9x - 9.9x safety factor)

Component	InternVL3-8B	InternVL3 %	Llama-3.2 -11B	Llama %
System Prompt	0	0%	0	0%
User Prompt	700	2.1%	700	0.5%
Image Tokens	3,000	9.2%	6,000	4.7%
History	0	0%	0	0%
Response	1,024	3.1%	4,096	3.2%
Total Used	**4,724**	**14.4%**	**10,796**	**8.4%**
Remaining	**28,044**	**85.6%**	**117,204**	**91.6%**

Table 12: Context window utilization by component

8 Summary: Context Window Utilization

8.1 By Component

8.2 Model Comparison

Metric	InternVL3-8B	Llama-3.2-11B	Winner
Context Window	32,768	128,000	Llama (3.9x)
Total Used	4,724	10,796	InternVL3 (efficient)
Utilization %	14.4%	8.4%	Llama (headroom)
Safety Factor	4.9x	9.9x	Llama (safer)
Image Efficiency	256 tokens/tile	6 tokens/pixel	InternVL3 (better)

Table 13: Model comparison summary

9 Key Takeaways

1. **Extremely Conservative Settings**

   Both models use only 8-14% of available context, providing massive safety margins:

   - InternVL3-8B: 85.6% unused
   - Llama-3.2-11B: 91.6% unused

2. **Images Are Relatively Cheap**

   Despite being multimodal, images consume a small portion of context:

   - InternVL3: 4.7% - 15.6% for images
   - Llama: 3.1% - 6.25% for images

3. **Stateless Design Is Optimal**

   Zero tokens allocated to conversation history:

   - Keeps usage constant across batch
   - Prevents context accumulation
   - Maintains predictable behavior
   - Perfect for independent document processing

4. **Significant Growth Potential**

   With 3.8x to 9.9x headroom, you could:

   - Increase response length 5-10x
   - Add conversation history (3-10 turns)
   - Process multiple images per request
   - Add detailed system prompts
   - Implement multi-turn refinement

5. **Provider Compatibility**

   Current settings work on most platforms:

   - Self-hosted: Full support
   - AWS Bedrock: Full support
   - Oracle Cloud: Full support
   - Azure: Llama exceeds 8K limit

6. **Production-Ready Configuration**

   Your settings align with official recommendations:

   - InternVL3: 1,024 tokens (matches official default)
   - Llama: 4,096 tokens (within recommended 2,048-4,096 range)
   - Both provide excellent safety margins
   - Well-suited for production deployment

10 Recommendations

10.1 Current Implementation: Optimal

No changes needed. Your configuration is:

- Conservative and safe
- Aligned with official guidelines
- Optimized for stateless document processing
- Production-ready with significant headroom

10.2 Future Enhancements (If Needed)

1. **Increase Response Length** (Low Priority)

   - Current: 1,024 / 4,096 tokens
   - Could increase: 2,048 / 8,192 tokens
   - Rationale: Handle more complex documents

2. **Add System Prompt** (Optional)

   - Cost: 200-500 tokens
   - Benefit: Consistent model behavior

- Trade-off: Minimal impact on context

3. **Multi-Turn Refinement** (Advanced)
   - Enable conversation history
   - Allow follow-up corrections
   - Cost: 4,724 / 10,796 tokens per turn
   - Rationale: Iterative extraction quality improvement
4. **Multiple Images** (Future Feature)
   - Process multi-page documents
   - Cost: Additional 3,000 / 6,000 tokens per image
   - Limit: 5-9 images (InternVL3), 14-19 images (Llama)

11 References

11.1 InternVL3 Official Documentation

- **InternVL3 Research Paper** (arXiv:2504.10479v1)
  - https://arxiv.org/html/2504.10479v1
  - Image tokenization details (256 tokens per 448×448 tile)
- **InternVL3 Deployment Guide**
  - https://internvl.readthedocs.io/en/latest/internvl3.0/deployment.html
  - Session length recommendations (8192/16384 tokens)
- **InternVL3 Quick Start**
  - https://internvl.readthedocs.io/en/latest/internvl3.0/quick_start.html
  - Official max_new_tokens=1024 configuration

11.2 Llama 3.2 Vision Official Documentation

- **Meta AI Blog** - "Llama 3.2: Revolutionizing edge AI and vision"
  - https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/
  - 128K context window announcement
- **Oracle Documentation** - Meta Llama 3.2 11B Vision
  - https://docs.oracle.com/en-us/iaas/Content/generative-ai/meta-llama-3-2-11b.htm
  - Image token measurements (512×512 = 1,610 tokens)
- **Microsoft Q&A** - Llama 3.2 Context Window Issues
  - https://learn.microsoft.com/en-us/answers/questions/2150702/
  - Azure 8K limitation discussion
- **AWS Blog** - Introducing Llama 3.2 in Amazon Bedrock
  - https://aws.amazon.com/blogs/aws/introducing-llama-3-2-models-from-meta-in-amazon-be
  - Full 128K support on AWS

11.3 Related Project Documentation

- `CONTEXT_ACCUMULATION_ANALYSIS.md` - Stateless vs stateful processing
- `WHY_V100_IS_POOR_FOR_VLM.md` - Hardware constraints
- `V100_FIX_GUIDE.md` - Memory optimization strategies
- `CLAUDE.md` - Project architecture

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